

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Built features and optimized a *Python* (*FastAPI*) search system on *Elasticsearch*, scaling to **600K+** documents
- Contributed to architecture and design decisions in an *Agile* team of **10 engineers** on an *AWS*-hosted platform
- Simplified evidence retrieval with a text-retrieval algorithm that validates URLs and filters outliers with **IQR**
- Optimized data preprocessing by flagging **30%** of evidence documents as identical with a de-duplication algorithm

Product Development Intern | Jetson

January 2025 – February 2025

- Communicated recommendations directly to the CEO in a **5-page report** on strategies to optimize user engagement
- Tested **3 experimental features** through **7 iterations of UI/UX testing** in the live app with **100K+** users
- Simplified the onboarding flow in *Figma*, cutting it from **12 to 5 pages**, and validated the change with **A/B testing**

Data Analytics Intern | STEM E

June 2024 – July 2024

- Collected, organized and analyzed **200+ TikTok posts** from **3 years** using *Python*, *Microsoft Excel* and *Power BI*
- Found through data analysis that “science fun facts” posts earned **69.3%** of views from just **7%** of all content
- Communicated a “fun facts” recommendation to the Executive Director, who adopted it, lifting engagement **24%**

PROJECTS

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Deployed a **RAG** agent (*Claude Sonnet 4.5*) using context engineering over session-scoped vector documents
- Structured papers in a relational *PostgreSQL* database on *Supabase* and queried them with *SQL* by field and year
- Built a web application whose *React Flow* interface graphs a paper’s outbound citations across **26 research fields**
- Ingested papers from *OpenAlex* and backfilled metadata from *Semantic Scholar* in a dual-API ETL pipeline

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Built a *Python FastAPI* back end that ingests water, waste and energy data from all **3 tools** for anomaly detection
- Applied **4 Random Forest models** to detect anomalies in live water, waste and energy data from all **3 tools**
- Integrated a *Gemma* Voice LLM into a *Next.js* dashboard that delivers actionable steps from the anomaly models
- Directed a team of 5 from ideation and development to the final presentation. **USAII Global Hackathon Finalist**

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Engineered an LLM application: a *JSON schema* that guides a *Claude Sonnet 4.6* agent to extract truck details
- Evaluated the agent on truck photos, refining the schema until **10 consecutive images** were right on all **3 features**
- Preprocessed **15,000** truck-listing images for data quality by de-duplicating listings and filtering non-USD prices
- Transformed truck photos into structured features (condition, brand, model, year) that feed the pricing model
- Modeled truck prices with a *quantile regression* on extracted features, scoring **87% accuracy** on **30 samples**
- Won **2nd Place at UIUC Founder’s Hackathon** leading team of **4** on pricing algorithm for Kamion (YC S22)

EcoBin | Research Paper | [Source Code](#)

March 2026 – May 2026

- Trained an *EfficientNetV2-S* model (**4.39M** parameters) in *PyTorch* on **15K+** images for **96.13% accuracy**
- Integrated a *PyTorch* classifier with an *Arduino Uno* smart bin prototype that sorts waste in **under 2 seconds**
- Built a synthetic data pipeline (*U2-Net* segmentation, alpha-compositing) producing **9K** contamination images
- Published the work as an *arXiv* paper with **2 datasets** on *Kaggle*. **2nd at Arizona State Science Fair**

SKILLS

Programming Languages: Python, Java, SQL, JavaScript, HTML/CSS, C++

Frameworks/Platforms: React, Next.js, Node.js, FastAPI, Flask, MySQL, PostgreSQL, AWS, Supabase, Firebase

AI/ML: scikit-learn, PyTorch, TensorFlow, Keras, Claude API, Hugging Face, OpenCV, Gemma 4, Vercel AI SDK

Concepts/Practices: Scalable Systems, Feature Engineering, Data Preprocessing, LLM Applications, RAG, Evals